

Functions and Function Notation (Part II)

Domain: All of the inputs of a function (typically x-values) that are able to produce an output.

Range: All of the outputs of a function (typically y-values) that are produced from the inputs in the domain.

✱ Neither imaginary nor complex numbers are allowed to be inputs or outputs of functions.

Finding Domain and Range with the Library of Functions

Example

Find the domain and range of the function below.

$$f(x) = \frac{1}{x}$$

Domain

Reasoning: A fraction with zero as its denominator is undefined, so x cannot be zero.

Words: All real numbers except zero.

Set Notation: $\{x | x \neq 0\}$

Interval Notation: $(-\infty, 0) \cup (0, \infty)$

Range

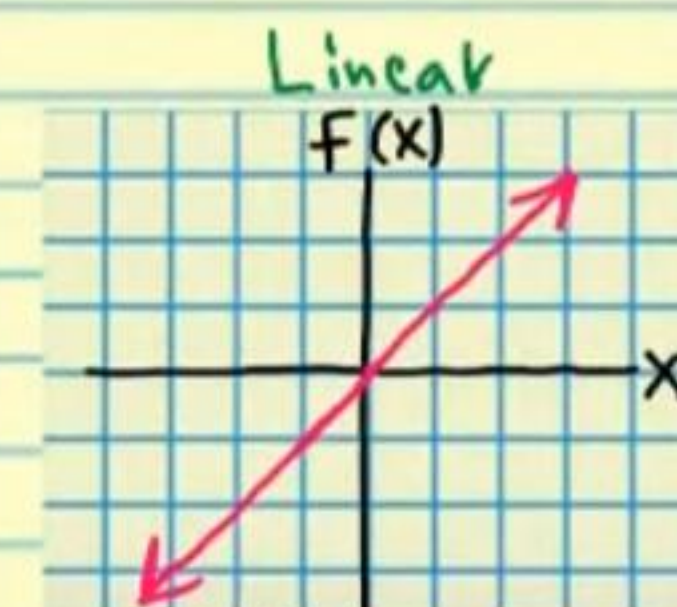
Words: All real numbers except zero.

Set Notation: $\{y | y \neq 0\}$

Interval Notation: $(-\infty, 0) \cup (0, \infty)$

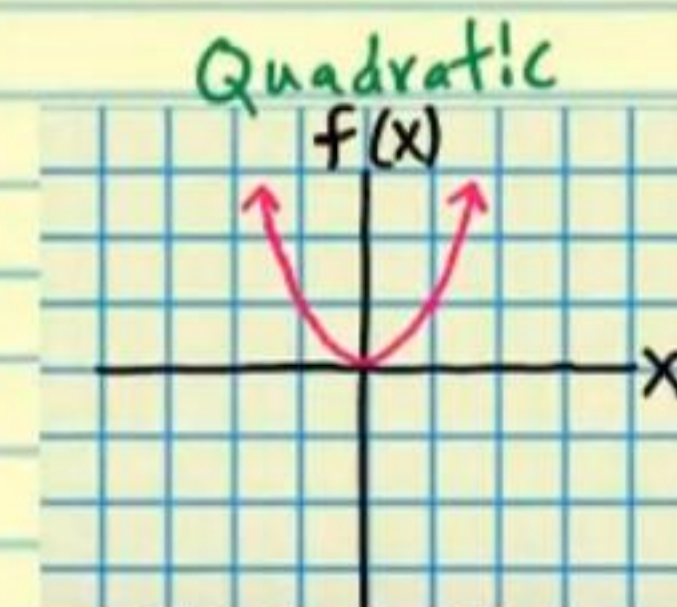
The Library of Functions

✱ Also called "parent" functions.



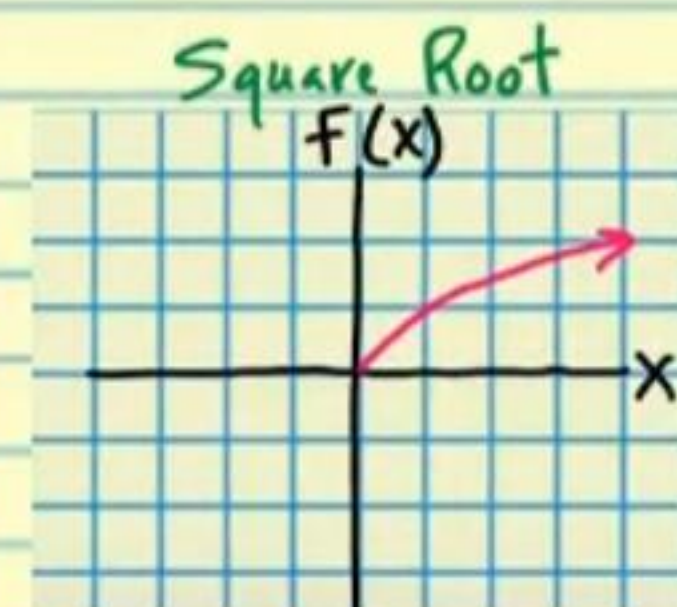
$$f(x) = x$$

x	-2	-1	0	1	2
f(x)	-2	-1	0	1	2



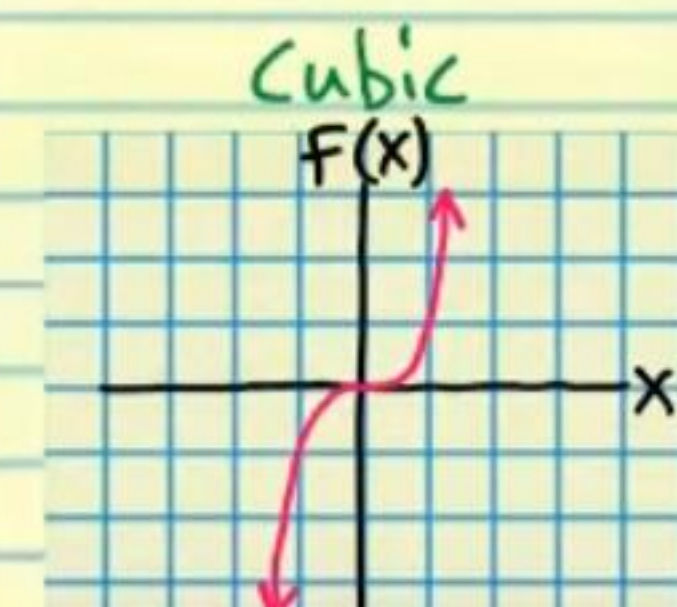
$$f(x) = x^2$$

x	-2	-1	0	1	2
f(x)	4	1	0	1	4



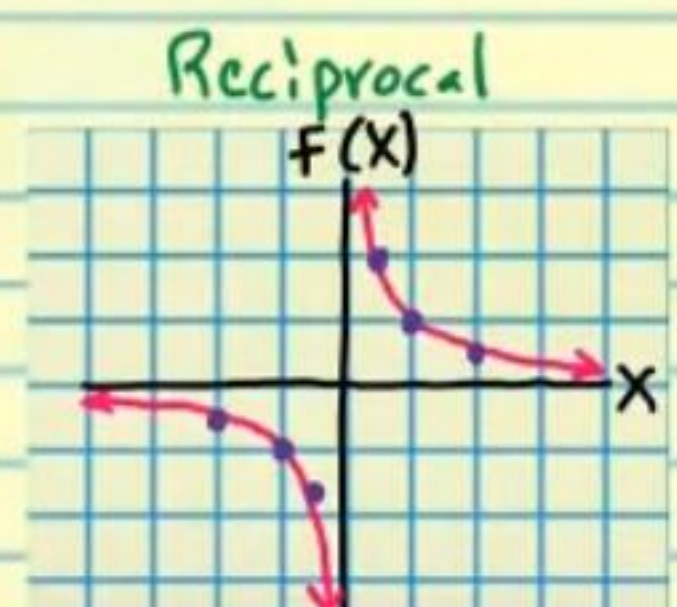
$$f(x) = \sqrt{x}$$

x	0	1	4	9	16
f(x)	0	1	2	3	4



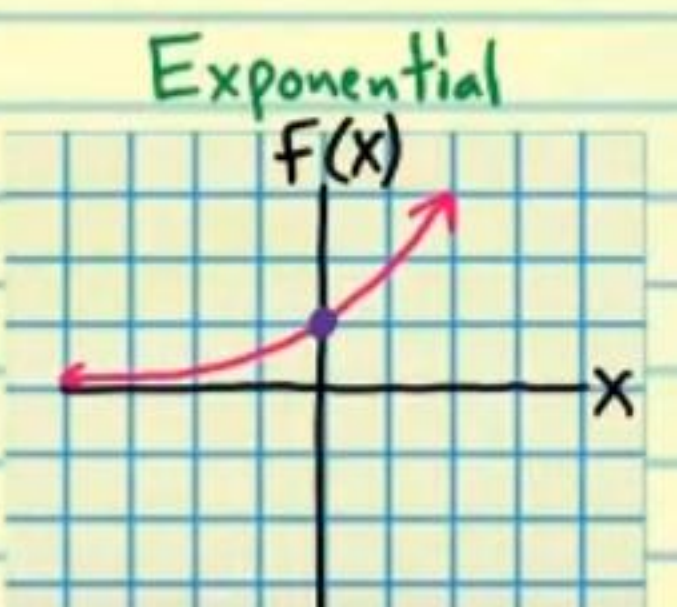
$$f(x) = x^3$$

x	-2	-1	0	1	2
f(x)	-8	-1	0	1	8



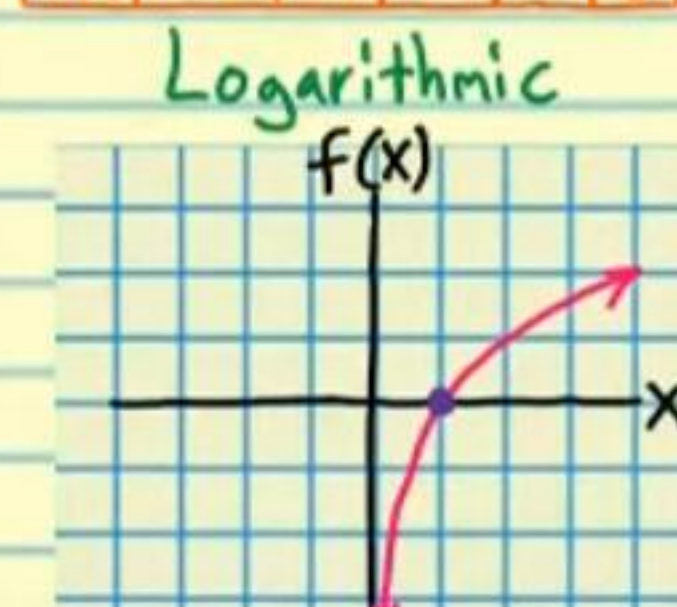
$$f(x) = \frac{1}{x}$$

x	-2	-1	-1/2	0	1/2	1	2
f(x)	-1/2	-1	-2	und	2	1	1/2



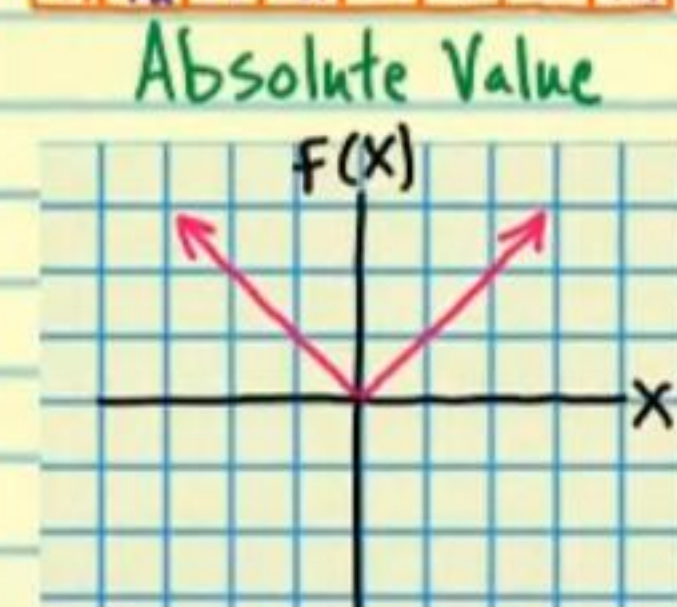
$$f(x) = a^x, a > 1$$

x	-	-	0	-	-
f(x)	-	-	1	-	-



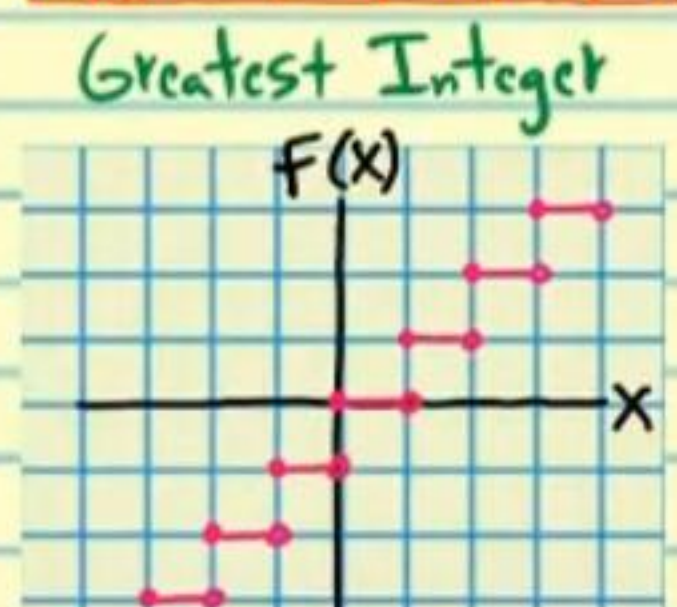
$$f(x) = \log_a x$$

x	0	1	-	-	-
f(x)	und	0	-	-	-



$$f(x) = |x|$$

x	-2	-1	0	1	2
f(x)	2	1	0	1	2



$$f(x) = [x]$$

x	-2 1/2	-1	0	1/2	1 1/2
f(x)	-3	-1	0	0	1

The four functions below are from the library. Use the graphs in the library to write the domain and range in words, set notation, and interval notation.

① $f(x) = |x|$ Domain

Words: All real numbers

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than or equal to zero.

Set Notation: $\{y | y \geq 0\}$

Interval Notation: $[0, \infty)$

② $f(x) = x^3$

Words: All real numbers. Domain

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

③ $f(x) = a^x, a > 1$

Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than zero.

Set Notation: $\{y | y > 0\}$

Interval Notation: $(0, \infty)$

④ $f(x) = \log_a x, a > 0$ and $a \neq 1$.

Domain

Words: All real numbers greater than zero.

Set Notation: $\{x | x > 0\}$

Interval Notation: $(0, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Finding the Domain and Range of More Complicated Functions

Graph the following functions and use the graphs to determine the domain and range. Write in word form, set notation, and interval notation.

⑤ $f(x) = \sqrt{-2x-3} - 2$

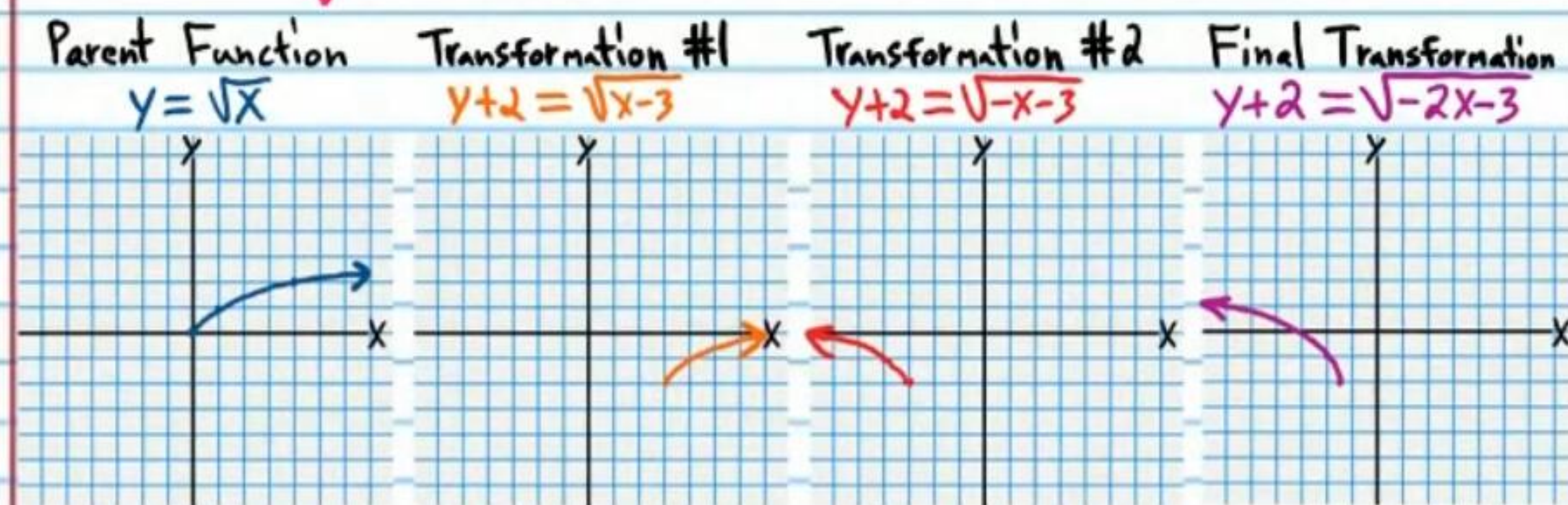
↓ Convert $f(x)$ to y .

$$y = \sqrt{-2x-3} - 2$$

↓ Change the form of the equation.

$$y+2 = \sqrt{-2x-3}$$

↓ Graph the equation.



Domain

Words: All real numbers less than or equal to -1.5.

Set Notation: $\{x | x \leq -1.5\}$

Interval Notation: $(-\infty, -1.5]$

Range

Words: All real numbers greater than or equal to -2.

Set Notation: $\{y | y \geq -2\}$

Interval Notation: $[-2, \infty)$

⑥ $f(x) = -3^{\frac{x}{3}+2}$

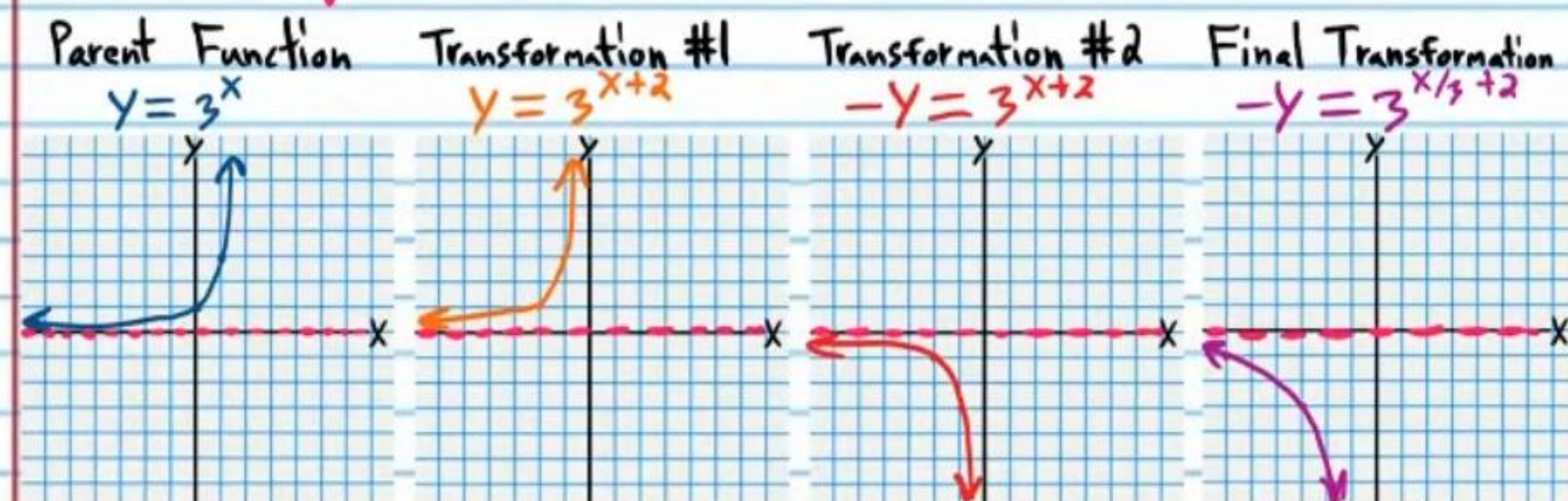
↓ Convert $f(x)$ to y .

$$y = -3^{\frac{x}{3}+2}$$

↓ Change the form of the equation.

$$-y = 3^{\frac{x}{3}+2}$$

↓ Graph the equation.



Domain

Words: All real numbers

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers less than zero.

Set Notation: $\{y | y < 0\}$

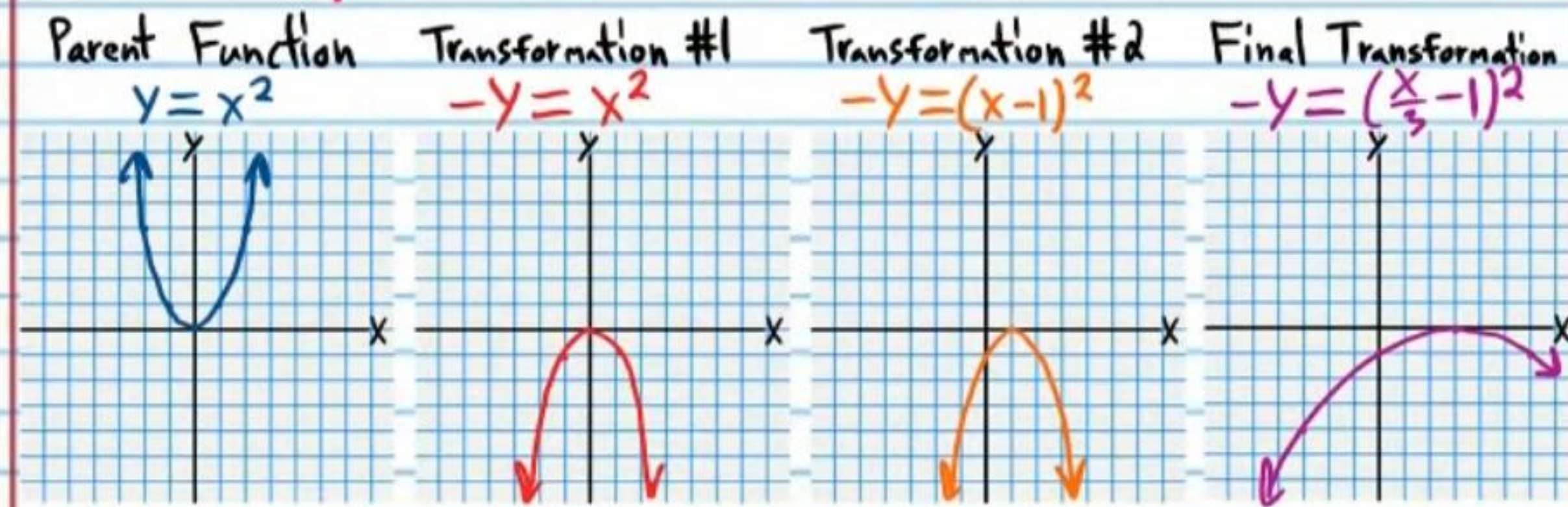
Interval Notation: $(-\infty, 0)$

⑦ $f(x) = -(\frac{x}{3}-1)^2$

↓ Convert $f(x)$ to y .
 $y = -(\frac{x}{3}-1)^2$

↓ Change the form of the equation.
 $-y = (\frac{x}{3}-1)^2$

↓ Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$
 Range

Words: All real numbers less than or equal to zero.

Set Notation: $\{y \mid y \leq 0\}$

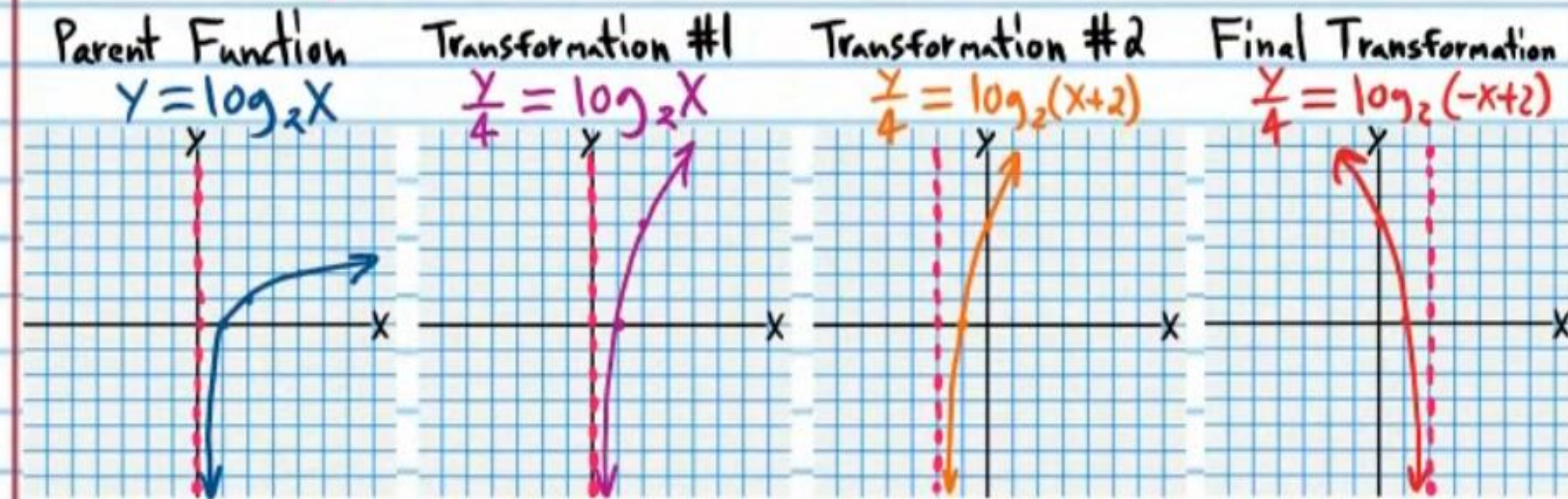
Interval Notation: $(-\infty, 0]$

⑧ $f(x) = 4\log_2(-x+2)$

↓ Convert $f(x)$ to y .
 $y = 4\log_2(-x+2)$

↓ Change the form of the equation.
 $\frac{y}{4} = \log_2(-x+2)$

↓ Graph the equation.



Domain

Words: All real numbers less than 2.

Set Notation: $\{x \mid x < 2\}$

Interval Notation: $(-\infty, 2)$
 Range

Words: All real numbers.

Set Notation: All real numbers.

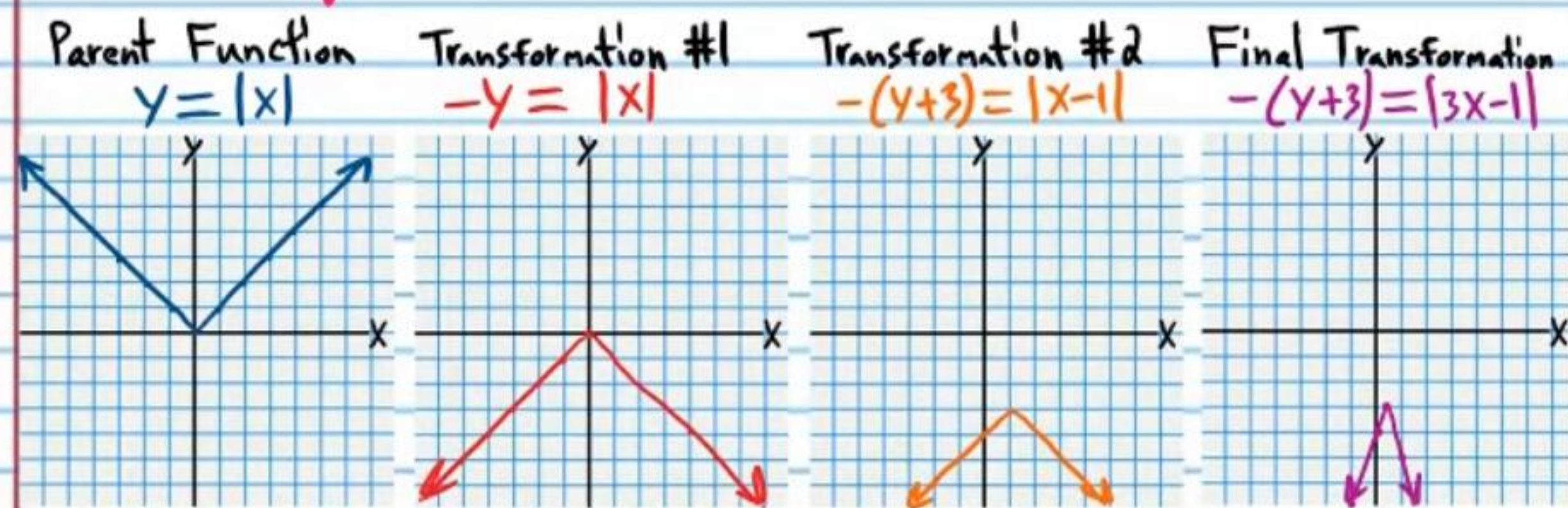
Interval Notation: $(-\infty, \infty)$

⑨ $F(x) = -|3x-1|-3$

↓ Convert $F(x)$ to y .
 $y = -|3x-1|-3$

↓ Change the form of the equation.
 $-(y+3) = |3x-1|$

↓ Graph the equation.



Domain
 Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$
 Range

Words: All real numbers less than or equal to -3.

Set Notation: $\{y | y \leq -3\}$

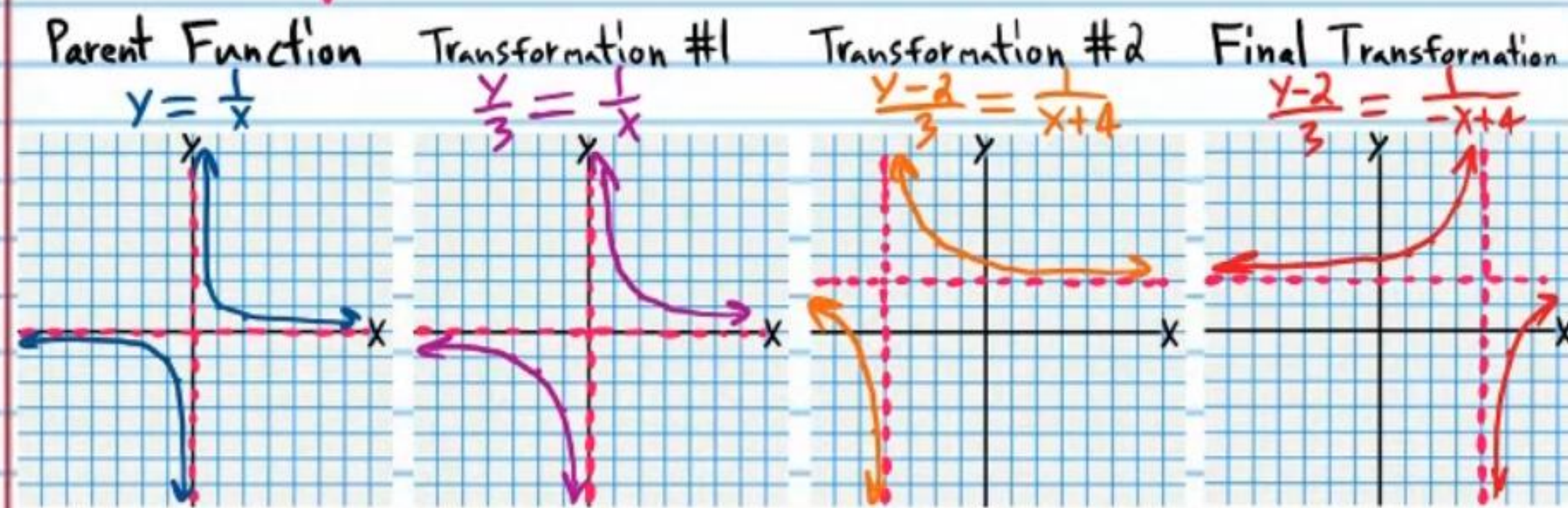
Interval Notation: $(-\infty, -3]$

⑩ $F(x) = \frac{3}{-x+4} + 2$

↓ Convert $F(x)$ to y .
 $y = \frac{3}{-x+4} + 2$

↓ Change the form of the equation.
 $\frac{y-2}{3} = \frac{1}{-x+4}$

↓ Graph the equation.



Domain
 Words: All real numbers except 4.

Set Notation: $\{x | x \neq 4\}$

Interval Notation: $(-\infty, 4) \cup (4, \infty)$
 Range

Words: All real numbers except 2.

Set Notation: $\{y | y \neq 2\}$

Interval Notation: $(-\infty, 2) \cup (2, \infty)$

$$\textcircled{11} f(x) = -\frac{(x+1)^3}{4}$$

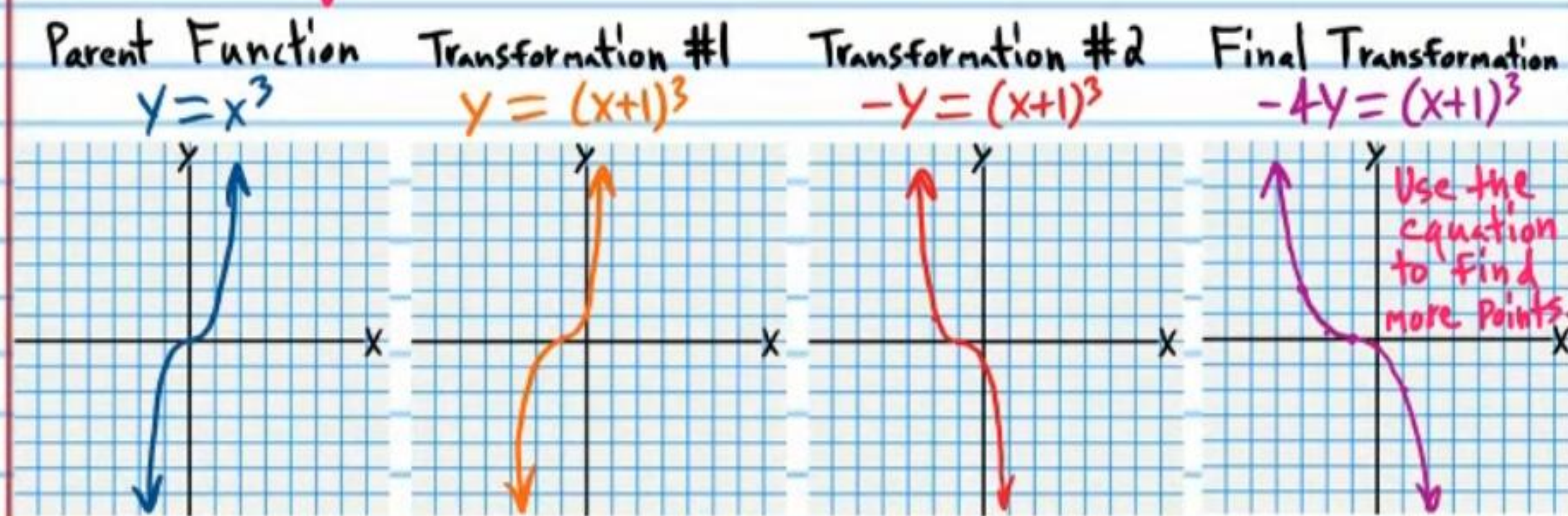
Convert $f(x)$ to y .

$$y = -\frac{(x+1)^3}{4}$$

Change the form of the equation.

$$-4y = (x+1)^3$$

Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

$$\textcircled{12} f(x) = \frac{2}{-x-2} - 1$$

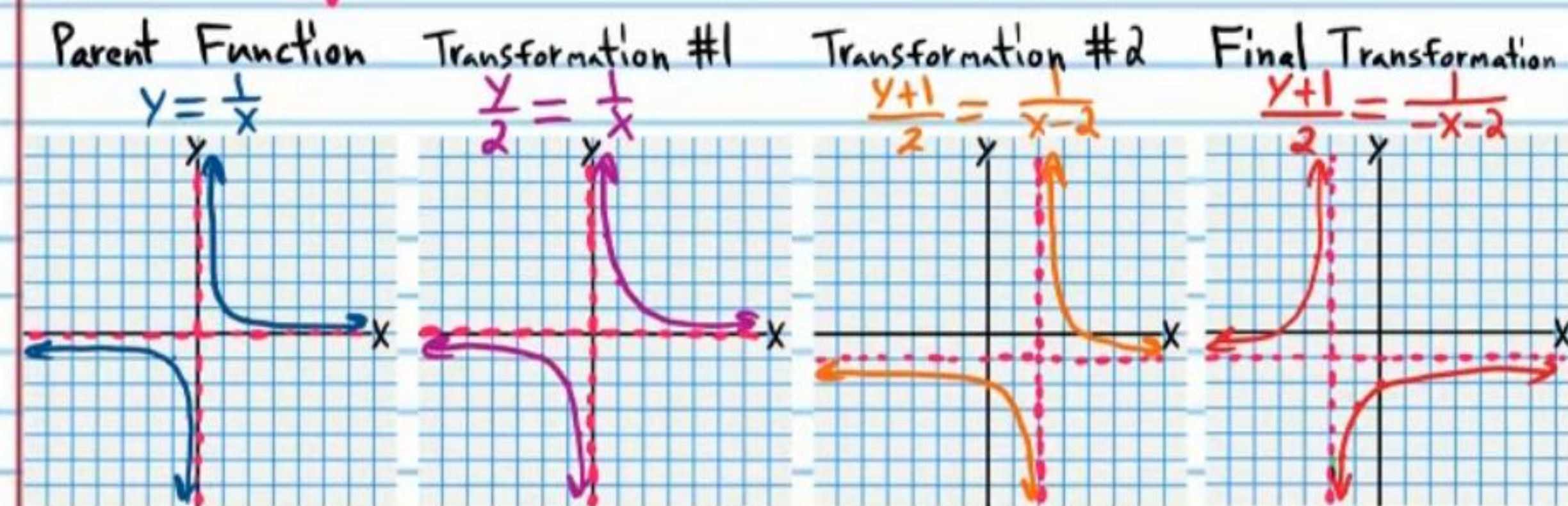
Convert $f(x)$ to y .

$$y = \frac{2}{-x-2} - 1$$

Change the form of the equation.

$$\frac{y+1}{2} = \frac{1}{-x-2}$$

Graph the equation.



Domain

Words: All real numbers except -2 .

Set Notation: $\{x \mid x \neq -2\}$

Interval Notation: $(-\infty, -2) \cup (-2, \infty)$

Range

Words: All real numbers except -1 .

Set Notation: $\{y \mid y \neq -1\}$

Interval Notation: $(-\infty, -1) \cup (-1, \infty)$

⑬ $F(x) = -2(x-4)^3 + 2$

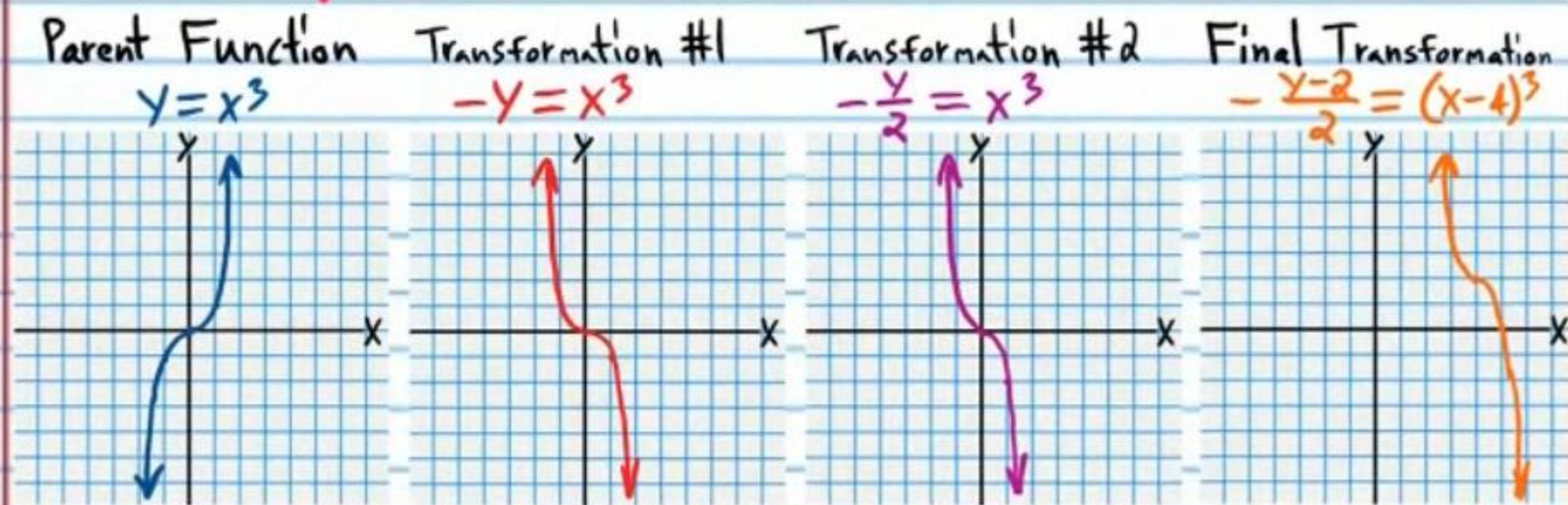
↓ Convert $F(x)$ to y .

$$y = -2(x-4)^3 + 2$$

↓ Change the form of the equation.

$$-\frac{y-2}{2} = (x-4)^3$$

↓ Graph the equation.



Words: All real numbers. Domain

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$ Range

Words: All real numbers.

Set Notation: All real numbers

Interval Notation: $(-\infty, \infty)$

⑭ $F(x) = -\frac{\log_4(x-2)}{2} - 3$

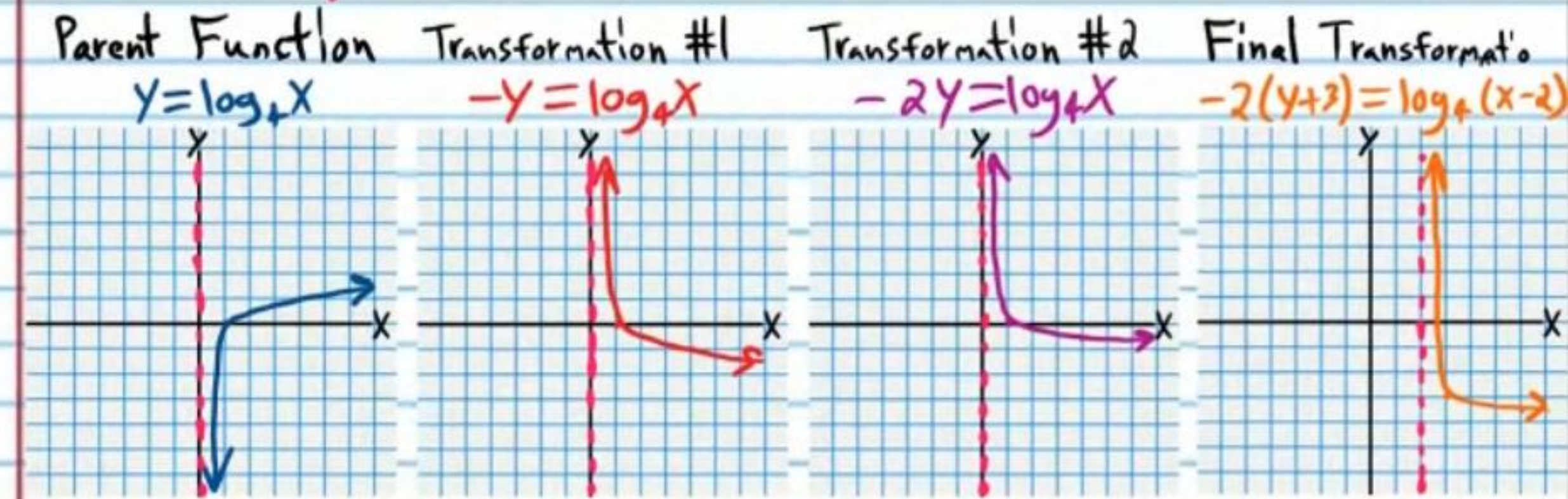
↓ Convert $F(x)$ to y .

$$y = -\frac{\log_4(x-2)}{2} - 3$$

↓ Change the form of the equation.

$$-2(y+3) = \log_4(x-2)$$

↓ Graph the equation.



Words: All real numbers greater than 2. Domain

Set Notation: $\{x | x > 2\}$

Interval Notation: $(2, \infty)$ Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

$$\textcircled{15} f(x) = -\frac{|x+3|}{3} - 2$$

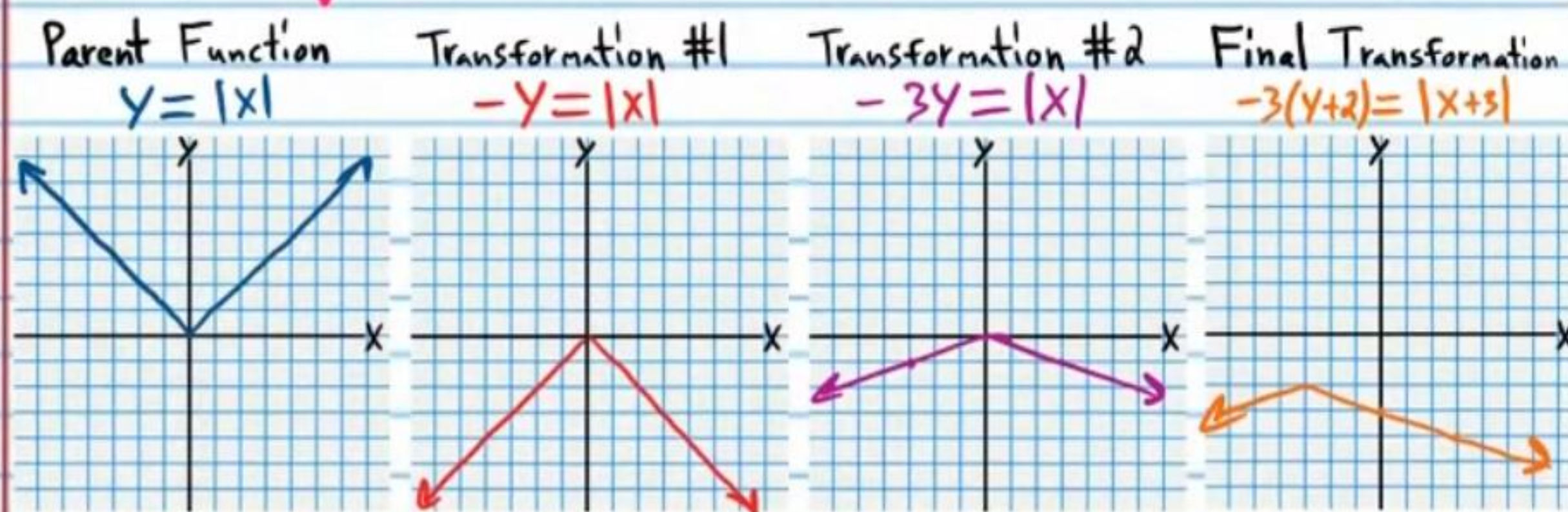
Convert $f(x)$ to y .

$$y = -\frac{|x+3|}{3} - 2$$

Change the form of the equation.

$$-3(y+2) = |x+3|$$

Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers less than or equal to -2 .

Set Notation: $\{y | y \leq -2\}$

Interval Notation: $(-\infty, -2]$

$$\textcircled{16} f(x) = -5(4^{x-2}) + 3$$

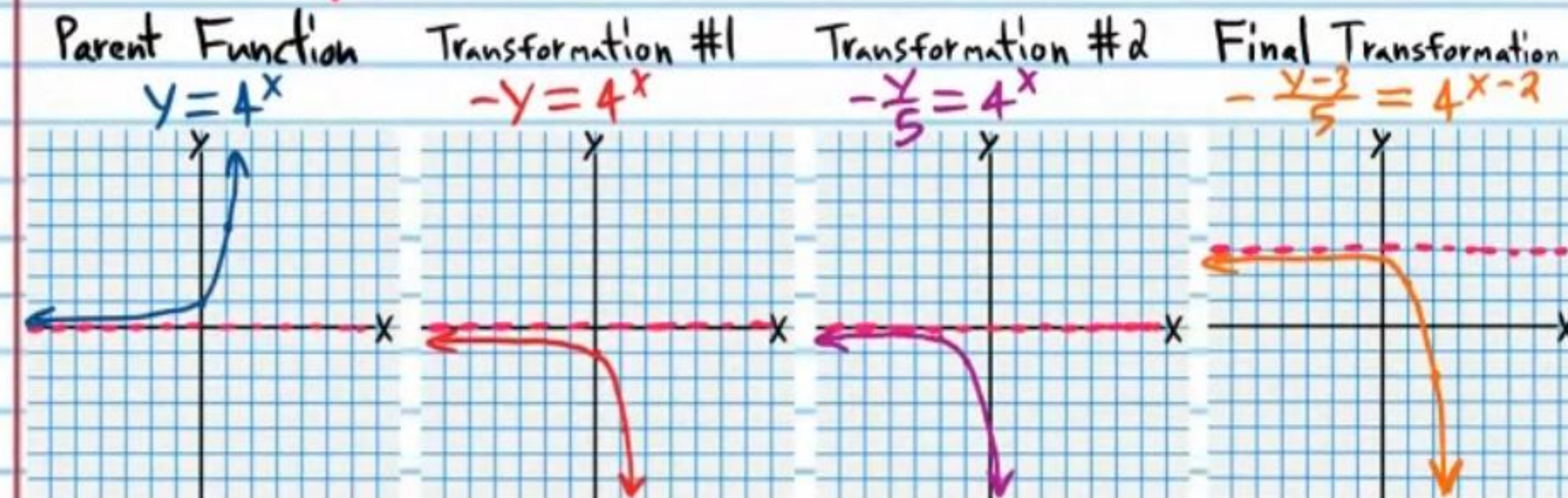
Convert $f(x)$ to y .

$$y = -5(4^{x-2}) + 3$$

Change the form of the equation.

$$-\frac{y-3}{5} = 4^{x-2}$$

Graph the equation.



Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers less than 3 .

Set Notation: $\{y | y < 3\}$

Interval Notation: $(-\infty, 3)$